SHIVAKUMAR THOTA

AI QA Lead | GenAI, LLM, RAG and Agentic AI Testing | AI Quality Engineering

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PROFESSIONAL SUMMARY

AI QA Lead with 12+ years of experience in quality engineering and test automation, now focused on Generative AI, LLM and Agentic AI testing at Mohawk Industries. Defines AI test strategy and quality standards for LLM applications built on Claude and OpenAI models, Retrieval-Augmented Generation (RAG) workflows and AI agents. Hands-on in prompt testing, AI response evaluation, hallucination testing, groundedness and context validation, and tool calling validation for agentic workflows. Uses a strong API and automation background with Java, Selenium, TOSCA and Postman to build AI regression suites and run AI integration and end-to-end AI validation. Leads AI QA work in Agile teams, working with developers, AI engineers and product owners on AI test planning, defect triage and release readiness.

AI AND TECHNICAL SKILLS

- **AI and GenAI Testing:** GenAI Testing, LLM Testing, Prompt Testing and Validation, AI Response Evaluation, Hallucination Testing, Groundedness Testing, Context Validation, Structured Output (JSON) Validation
- **RAG Testing:** Retrieval-Augmented Generation (RAG) Validation, Document Ingestion Testing, Retrieval Accuracy, Context Relevance, Source Grounding, Citation Validation, Semantic Search Testing
- **Agentic AI Testing:** AI Agent Testing, Multi-Agent Workflow Testing, Tool Calling Validation, Agent Decision Validation, State and Context Validation, Human-in-the-Loop Workflows, Failure and Retry Scenarios
- **AI Quality Engineering:** AI Test Strategy, AI Test Planning, Risk-Based Testing, AI Functional Testing, AI Integration Testing, AI Regression Testing, End-to-End AI Validation
- **LLMs and AI Platforms:** Claude (Anthropic), OpenAI GPT Models, LLM APIs, Enterprise Knowledge Bases, Vector Search
- **API and Automation:** REST API Testing, Postman, SoapUI, Swagger, Selenium WebDriver, TOSCA, Java, Cucumber (BDD), TestNG, JUnit, Maven, SQL
- **Test Management:** Jira, qTest, Confluence, Defect Management, Test Planning, Quality Metrics, Release Validation
- **Enterprise Platforms:** SAP C4C, Salesforce, SAP Commerce (Hybris), ESB and Mainframe Integrations
- **DevOps and Version Control:** Jenkins, Bamboo, Git, GitHub
- **Methodologies:** Agile, Scrum

PROFESSIONAL EXPERIENCE

Mohawk Industries, Calhoun, GA

QA Lead | July 2020 to Present

Leads quality engineering for AI-enabled enterprise applications built on Claude and OpenAI models, including GenAI assistants, RAG-based knowledge search and AI agent workflows that support business users.

AI QA Leadership and Strategy

- Own the test strategy, quality standards and release criteria for AI-enabled enterprise applications, including GenAI and LLM-based solutions.
- Review AI requirements and solution designs with product owners and architects to raise quality risks such as hallucination, missing context and unclear expected behavior early in the lifecycle.
- Plan AI testing across functional, API, integration, regression and end-to-end levels, covering estimation, sprint planning and work allocation across the QA team.
- Run defect triage for AI issues with developers and AI engineers, separating prompt, retrieval, model and application defects so each reaches the right owner.

- Provide test coverage, quality metrics, open risks and release readiness updates to project leads and business stakeholders.
- Mentor QA engineers on AI testing methods, including prompt design, response evaluation and handling non-deterministic outputs.

GenAI, LLM and RAG Testing

- Designed prompt test sets for LLM applications covering normal, negative, ambiguous and edge-case inputs, which widened AI test coverage beyond the expected-path scenarios.
- Compare and evaluate Claude and OpenAI model responses for accuracy, relevance, completeness, consistency, instruction following, business rule compliance and response format, including JSON outputs used by downstream systems.
- Test hallucination and out-of-scope scenarios to confirm the application declines or flags questions it cannot answer from its data.
- Validate RAG workflows by checking document ingestion, retrieved chunks, context relevance, source grounding and the accuracy of the final answer.
- Test retrieval against query variations, missing context and conflicting information across enterprise knowledge sources, and confirm that citations point to the correct documents.

AI Agent, API and Integration Testing

- Validate AI agent workflows across multi-step business processes, covering tool selection, API calls, context retention, state handling and the final output.
- Test agent failure paths such as incorrect tool selection, incomplete execution, timeouts, retry logic and handoffs to human reviewers.
- Test AI service APIs and their integrations, checking request and response payloads, authentication, JSON structure, error handling and data flow into connected enterprise systems.

AI Regression and Automation

- Created repeatable AI regression scenarios that compare response quality after prompt, model, configuration, knowledge base and retrieval changes, using defined quality criteria instead of exact-text matching.
- Automated repeatable AI checks through dataset-driven API tests and reusable response validations, built on the existing Java and API automation framework and run through the CI/CD pipeline.
- Identified hallucination, grounding and context-handling defects before production releases and worked with engineering teams to analyze inconsistent LLM responses.

Enterprise Application Testing

- Built and maintained TOSCA automation suites for sanity, minimal and full regression across SAP C4C, Salesforce and SAP Commerce (Hybris), covering Accounts, Contacts, Leads, Opportunities, Sales Quotes and Sales Orders.
- Validated data flow between these platforms and the Mainframe (CAMS) and ESB integrations, using qTest for test execution, Jira for defect tracking and Bamboo for scheduled regression runs.

Environment: Agile/Scrum, Claude, OpenAI GPT Models, RAG, AI Agents, REST APIs, Postman, TOSCA, SAP C4C, Salesforce, SAP Commerce (Hybris), Jira, qTest, Confluence, Bamboo, Git

World Fuel Services Corporation, Miami, FL (through Collaborate Solutions, Inc.)
QA Automation Engineer | April 2018 to June 2020

- Designed and maintained Selenium WebDriver automation in Java using the Page Object Model, with data-driven, keyword-driven and hybrid framework components.
- Wrote BDD scenarios and step definitions in Cucumber (Gherkin), and ran and reported tests through TestNG and Maven.
- Tested SOAP and REST web services with Postman, SoapUI and Swagger, adding assertions to validate request and response payloads.

- Ran automated regression on each build through Jenkins, with GitHub for source control.
- Wrote SQL queries to prepare test data and verify backend results.
- Performed functional, integration, regression and cross-browser testing, including parallel runs on Selenium Grid.
- Took part in sprint planning, estimation and daily stand-ups, and prepared test summary reports after each test cycle.

EDUCATION

- Master's in Computer Science, Rivier University, Nashua, NH, USA, 2018
- Bachelor's in Computer Science, Jawaharlal Nehru Technological University, Hyderabad, India, 2013

CERTIFICATIONS

- Tricentis Tosca: Automation Specialist Level 1 and 2 (AS1, AS2), Automation Engineer Level 1 (AE1), Test Design Specialist Level 1 and 2 (TDS1, TDS2), Tosca Query Language (TQL), Exploratory Testing Specialist
- Certified Automation Specialist for SAP